

Quantum Gravitation

Problem Set 2

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Topic

Normal Modes of the Gravitational Medium

Submission

Submit one PDF. Show the derivation, identify every approximation, and include a short consistency check. The maximum file size is 25 MB.

Problems

1. Derive the first displayed relation used in Normal Modes of the Gravitational Medium. State all assumptions and conventions.

$$h_{ij}(x) = \sum_{k,s} q_{ks} e_{ij}^s(k) e^{ikx}$$

2. Calculate a nontrivial case using the second relation. Carry units and uncertainty where applicable.

$$H = \frac{1}{2} \sum_{k,s} (p_{ks}^2 + \omega_{ks}^2 q_{ks}^2)$$

3. Verify the third relation in two limiting regimes and explain the physical meaning of the result.

$$\mathcal{E}_{ij,kl} e_{kl}^s = \rho_m \omega_s^2 e_{ij}^s$$

Show enough intermediate work that another student can reproduce the calculation.